

Categorical variables. Visualization techniques

Ice cream shop

Note: You may solve these problems both on paper or in a software of your choice. The medium is not crucial.

Background There is an ice cream shop, that is operating in New York, LA and San Francisco.

Data You have sold 12,327 ice creams in New York; 17,129 in LA and 19,923 in San Francisco.

Task Order the data in a frequency distribution table.

Solution:

	<u>Frequency</u>
New York	12,327
LA	17,129
San Francisco	19,923
<u>Total</u>	<u>49,379</u>

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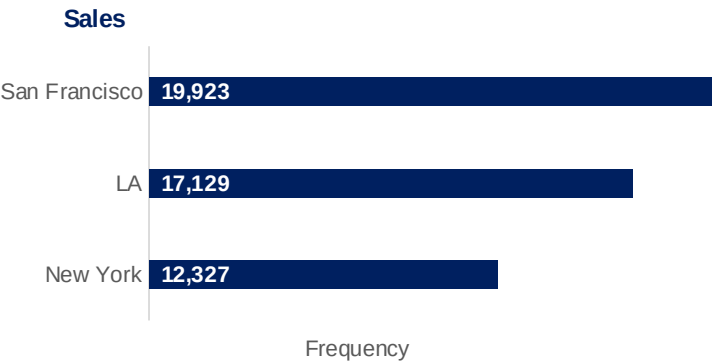
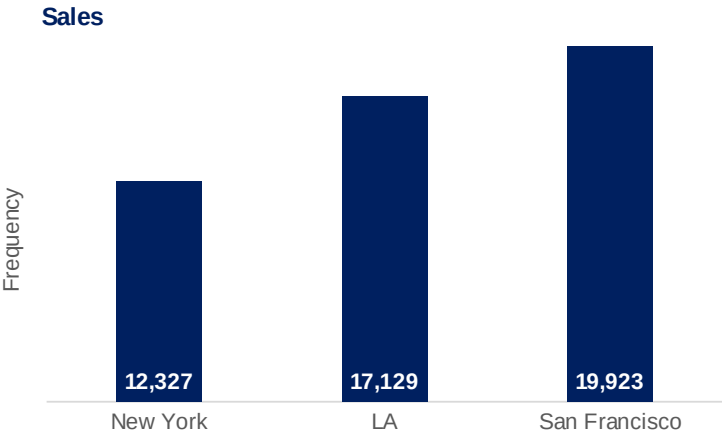
Ice cream shop

Background You have a frequency distribution table with all the sales.
Task Create a bar chart, representing the data. Note that in Excel, the term for a bar charts are called both: 'column charts' and 'bar charts'.

Solution:

	Frequency
New York	12,327
LA	17,129
San Francisco	19,923
Total	49,379

In order to create a chart in excel, you should select the data. In this case the cells B11, B12, B13, C11, C12 and C13. Then you click Insert and then choose the appropriate chart for your case. Both solutions are acceptable. However, the left one is the one that you would observe more often.



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- Background**
- You have a frequency distribution table with all the sales.
- Task 1**
- Create a new column in your table, representing the relative frequency of input. You can choose to express it in percentages or as a decimal.
- Task 2**
- Create a pie chart, representing the share of each city to the sales of your company.

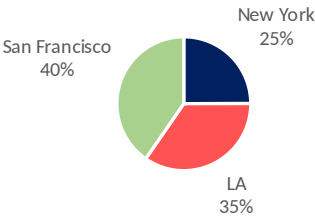
Solution:

	Frequency	Relative frequency	or.. relative frequency
New York	12,327	25%	0.25
LA	17,129	35%	0.35
San Francisco	19,923	40%	0.40
Total	49,379	100%	1.00

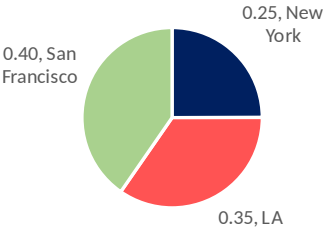
You may choose to create your pie chart using the frequency, relative frequency in % or in decimals.
All three solutions will result in the same pie chart.
Generally, you should use the absolute frequency, but put the relative one as labels for the chart.
Excel gives the possibility to put the relative frequency, stated in % as data labels, even if you haven't calculated it.

In the left chart, we have used the automatic labeling.
In the right chart, we have put the labels as cell values from the table.

Ice cream sales share



Ice cream sales share



Categorical variables. Visualization techniques

Ice cream shop

Background You have a frequency distribution table with all the sales. You also have the relative frequency from the pie chart problem.

Task 1 Order the table by frequency.

Task 2 Create a bar (column) chart representing the ordered data.

Task 3 In a new column, calculate the cumulative frequency of the data.

Task 4 On a second axis in the same chart, represent the cumulative frequency of the data.

Solution:

<i>Ordered</i>	Frequency	Relative frequency	Cumulative frequency
San Francisco	19,923	40%	40%
LA	17,129	35%	75%
New York	12,327	25%	100%
Total	49,379	100%	

Adding a second axis is not so straightforward in Excel.

This may be done in various ways. Here is a link to Microsoft's article on the topic. [Link](#)

For the purposes of statistics, you need to understand the application of the cumulative frequency line.

Drawing it in Excel is not top priority for this course.

